

Weekly Meeting - Juil Koo

Denoising Diffusion Implicit Models

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Slide Template Credit: Charlie

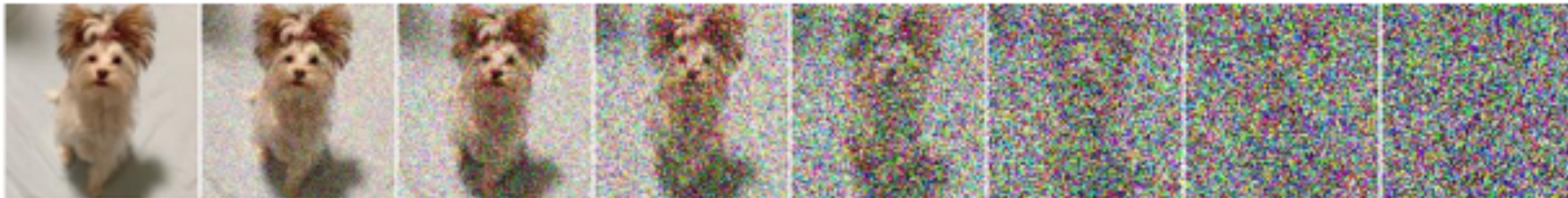
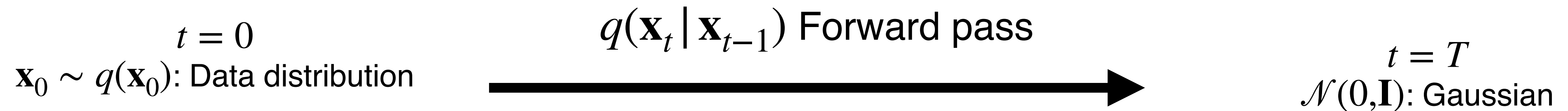


Background: DDPM

Let's define a **Markov chain** between data distribution $q(\mathbf{X}_0)$ and $\mathcal{N}(0, \mathbf{I})$ such that

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I}).$$

← Gaussian!



Background: DDPM

Gaussian has so many nice properties!

- Forward pass: $q(\mathbf{x}_t | \mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$

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- Forward pass: $q(\mathbf{x}_t | \mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$
- Sampling $\mathbf{x}_t$ at an arbitrary t : $q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$,
where $\alpha_t := 1 - \beta_t$, $\bar{\alpha}_t := \prod_{s=1}^t \alpha_s$

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- Forward pass: $q(\mathbf{x}_t | \mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$
- Sampling $\mathbf{x}_t$ at an arbitrary t : $q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$,
where $\alpha_t := 1 - \beta_t$, $\bar{\alpha}_t := \prod_{s=1}^t \alpha_s$
- Reverse pass: $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) \tilde{\beta}_t \mathbf{I})$,
where $\tilde{\mu}_t := \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0 + \frac{\sqrt{\alpha_t} (1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t$ and $\tilde{\beta}_t := \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \beta_t$

Background: DDPM

Derive $q(\mathbf{x}_t | \mathbf{x}_0)$:

$$q(\mathbf{x}_t | \mathbf{x}_{t-1}) = \mathcal{N}(\mathbf{x}_t; \sqrt{\alpha_t} \mathbf{x}_{t-1}, (1 - \alpha_t) \mathbf{I})$$

$$\mathbf{x}_t = \sqrt{\alpha_t} \mathbf{x}_{t-1} + \sqrt{1 - \alpha_t} \epsilon_{t-1}$$

$$= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{\alpha_t (1 - \alpha_{t-1})} \epsilon_{t-2} + \sqrt{1 - \alpha_t} \epsilon_{t-1}$$

$$= \sqrt{\alpha_t \alpha_{t-1}} \mathbf{x}_{t-2} + \sqrt{1 - \alpha_t \alpha_{t-1}} \bar{\epsilon}_{t-2}$$

$$= \dots$$

$$= \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon$$

$$q(\mathbf{x}_t | \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I}), \text{ where } \bar{\alpha}_t = \prod_{s=1}^t \alpha_s.$$

$X \sim \mathcal{N}(0, \sigma_1^2 \mathbf{I})$, and $Y \sim \mathcal{N}(0, \sigma_2^2 \mathbf{I})$

$$\begin{aligned} \text{Var}(X + Y) &= \mathbb{E}[(X + Y)^2] - \mathbb{E}[X + Y]^2 \\ &= \mathbb{E}[X^2] + \mathbb{E}[Y^2] - 2\mathbb{E}[X]\mathbb{E}[Y] \\ &= \sigma_1^2 + \sigma_2^2 \end{aligned}$$

$\Rightarrow Z := X + Y, \quad Z \sim \mathcal{N}(0, (\sigma_1^2 + \sigma_2^2) \mathbf{I})$

Background: DDPM

Why the reverse pass $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$ is conditioned on both $\mathbf{x}_t$ and $\mathbf{x}_0$?

$$q(\mathbf{x}_{t-1} | \mathbf{x}_t) = q(\mathbf{x}_t | \mathbf{x}_{t-1}) \cdot \boxed{\frac{q(\mathbf{x}_{t-1})}{q(\mathbf{x}_t)}} \quad \text{unknown}$$



$$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) := q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0) \cdot \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_0)}{q(\mathbf{x}_t | \mathbf{x}_0)}$$

Background: DDPM

Derivation of $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I})$

$$\begin{aligned} q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) &= q(\mathbf{x}_t | \mathbf{x}_{t-1}, \mathbf{x}_0) \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_0)}{q(\mathbf{x}_t | \mathbf{x}_0)} \\ &\propto \exp\left(-\frac{1}{2}\left(\frac{(\mathbf{x}_t - \sqrt{\alpha_t} \mathbf{x}_{t-1})^2}{\beta_t} + \frac{(\mathbf{x}_{t-1} - \sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0)^2}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0)^2}{1 - \bar{\alpha}_t}\right)\right) \\ &= \exp\left(-\frac{1}{2}\left(\frac{\mathbf{x}_t^2 - 2\sqrt{\alpha_t} \mathbf{x}_t \mathbf{x}_{t-1} + \alpha_t \mathbf{x}_{t-1}^2}{\beta_t} + \frac{\mathbf{x}_{t-1}^2 - 2\sqrt{\bar{\alpha}_{t-1}} \mathbf{x}_0 \mathbf{x}_{t-1} + \bar{\alpha}_{t-1} \mathbf{x}_0^2}{1 - \bar{\alpha}_{t-1}} - \frac{(\mathbf{x}_t - \sqrt{\bar{\alpha}_t} \mathbf{x}_0)^2}{1 - \bar{\alpha}_t}\right)\right) \\ &= \exp\left(-\frac{1}{2}\left(\left(\frac{\alpha_t}{\beta_t} + \frac{1}{1 - \bar{\alpha}_{t-1}}\right) \mathbf{x}_{t-1}^2 - \left(\frac{2\sqrt{\alpha_t}}{\beta_t} \mathbf{x}_t + \frac{2\sqrt{\bar{\alpha}_{t-1}}}{1 - \bar{\alpha}_{t-1}} \mathbf{x}_0\right) \mathbf{x}_{t-1} + C(\mathbf{x}_t, \mathbf{x}_0)\right)\right) \\ &\quad \vdots \\ &\Rightarrow \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\alpha_t}(1 - \bar{\alpha}_{t-1})}{1 - \bar{\alpha}_t} \mathbf{x}_t + \frac{\sqrt{\bar{\alpha}_{t-1}} \beta_t}{1 - \bar{\alpha}_t} \mathbf{x}_0, \quad \tilde{\beta}_t = \frac{1 - \bar{\alpha}_{t-1}}{1 - \bar{\alpha}_t} \cdot \beta_t \end{aligned}$$

Background: DDPM

- Forward pass: $q(\mathbf{x} | \mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$
- Reverse pass: $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I})$

$t = 0$
 $\mathbf{x}_0 \sim q(\mathbf{x}_0)$: Data distribution

$q(\mathbf{x}_t | \mathbf{x}_{t-1})$ Forward pass

$t = T$
 $\mathcal{N}(\mathbf{0}, \mathbf{I})$: Gaussian

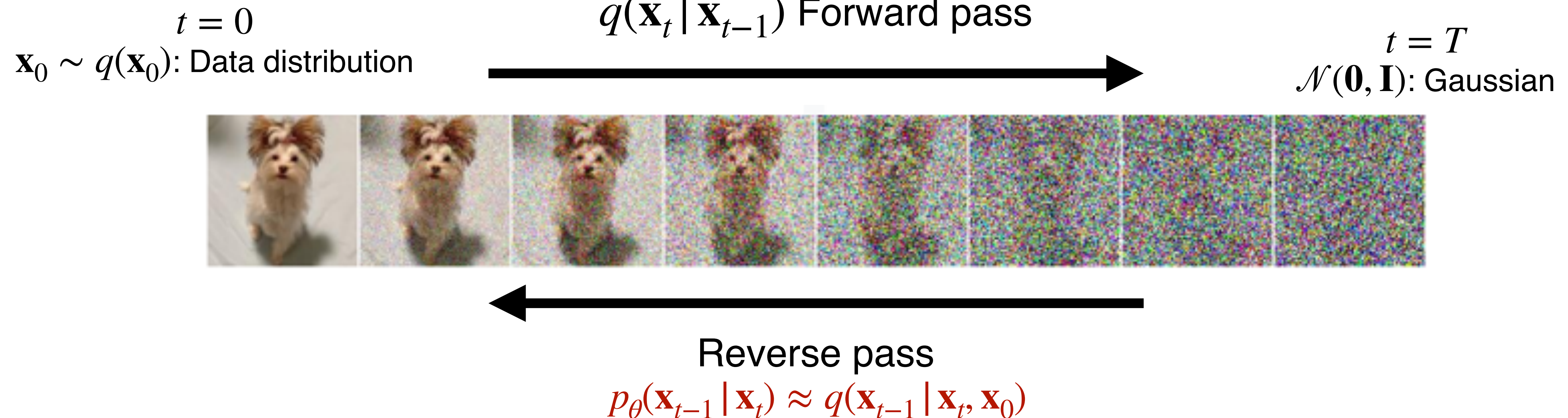


$q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)$ Reverse pass

Background: DDPM

Approximate the reverse process using parameter θ .

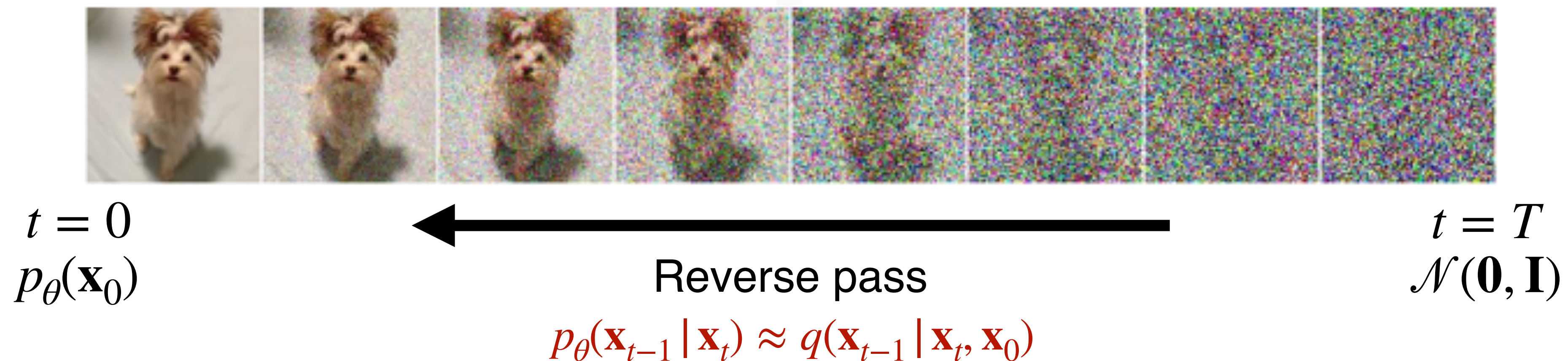
- Forward pass: $q(\mathbf{x} | \mathbf{x}_{t-1}) := \mathcal{N}(\mathbf{x}_t; \sqrt{1 - \beta_t} \mathbf{x}_{t-1}, \beta_t \mathbf{I})$
- Reverse pass: $q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}) \implies p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \tilde{\beta}_t \mathbf{I})$



Background: DDPM

Goal: Maximize $\log p_{\theta}(\mathbf{x}_0)$

$$\text{minimize } \mathbb{E}_{q(\mathbf{x}_0)}[-\log p_{\theta}(\mathbf{x}_0)] = \text{minimize } \mathbb{E}_q\left[\sum_t D_{KL}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) || p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t))\right]$$



Background: DDPM

Use the Evidence Lower Bound (ELBO):

Definition: $p_{\theta}(\mathbf{x}_{0:T}) := p_{\theta}(\mathbf{x}_T) \prod_{t=1}^T p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t)$, $q(\mathbf{x}_{1:T} | \mathbf{x}_0) := \prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1})$

$$\begin{aligned} -\log p_{\theta}(\mathbf{x}_0) &\leq -\log p_{\theta}(\mathbf{x}_0) + D_{KL}(q(\mathbf{x}_{1:T} | \mathbf{x}_0) \| p_{\theta}(\mathbf{x}_{1:T} | \mathbf{x}_0)) \\ &= -\log p_{\theta}(\mathbf{x}_0) + \mathbb{E}_{\mathbf{x}_{1:T} \sim q(\mathbf{x}_{1:T} | \mathbf{x}_0)} \left[\log \frac{q(\mathbf{x}_{1:T} | \mathbf{x}_0)}{p_{\theta}(\mathbf{x}_{0:T}) / p_{\theta}(\mathbf{x}_0)} \right] \\ &= \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_{1:T} | \mathbf{x}_0)}{p_{\theta}(\mathbf{x}_{0:T})} \right] . \end{aligned}$$

$$\text{Let } L_{VLB} = \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_{1:T} | \mathbf{x}_0)}{p_{\theta}(\mathbf{x}_{0:T})} \right] .$$

Background: DDPM

$$\begin{aligned} L_{VLB} &= \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_{1:T} | \mathbf{x}_0)}{p_\theta(\mathbf{x}_{0:T})} \right] \\ &= \mathbb{E}_q \left[\log \frac{\prod_{t=1}^T q(\mathbf{x}_t | \mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_T) \prod_{t=1}^T p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=1}^T \log \frac{q(\mathbf{x}_t | \mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \right] \\ &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_t | \mathbf{x}_{t-1})}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} + \log \frac{q(\mathbf{x}_t | \mathbf{x}_0)}{p_\theta(\mathbf{x}_0 | \mathbf{x}_1)} \right] \\ &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \left(\frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} \cdot \frac{q(\mathbf{x}_t | \mathbf{x}_0)}{q(\mathbf{x}_{t-1} | \mathbf{x}_0)} \right) + \log \frac{q(\mathbf{x}_t | \mathbf{x}_0)}{p_\theta(\mathbf{x}_0 | \mathbf{x}_1)} \right] \\ &= \mathbb{E}_q \left[-\log p_\theta(\mathbf{x}_T) + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} + \sum_{t=2}^T \frac{q(\mathbf{x}_t | \mathbf{x}_0)}{q(\mathbf{x}_{t-1} | \mathbf{x}_0)} + \log \frac{q(\mathbf{x}_t | \mathbf{x}_0)}{p_\theta(\mathbf{x}_0 | \mathbf{x}_1)} \right] \\ &= \mathbb{E}_q \left[\log \frac{q(\mathbf{x}_T | \mathbf{x}_0)}{p_\theta(\mathbf{x}_T)} + \sum_{t=2}^T \log \frac{q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0)}{p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)} - \log p_\theta(\mathbf{x}_0 | \mathbf{x}_1) \right] \\ &= \mathbb{E}_q \left[D_{KL}(q(\mathbf{x}_T | \mathbf{x}_0) \| p_\theta(\mathbf{x}_T)) + \sum_{t=2}^T D_{KL}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0 | \mathbf{x}_1) \right] \end{aligned}$$

Background: DDPM

$$\mathbb{E}_q[D_{KL}(q(\mathbf{x}_T | \mathbf{x}_0) \| p_\theta(\mathbf{x}_T)) + \sum_{t=2}^T D_{KL}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0 | \mathbf{x}_1)]$$

what we're interested in

$$\Rightarrow q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}) \quad \overset{\text{KL divergence}}{\longleftrightarrow} \quad p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \tilde{\beta}_t \mathbf{I})$$

Background: DDPM

$$\mathbb{E}_q[D_{KL}(q(\mathbf{x}_T | \mathbf{x}_0) \| p_\theta(\mathbf{x}_T)) + \sum_{t=2}^T D_{KL}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0 | \mathbf{x}_1)]$$

what we're interested in

$$\Rightarrow q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}) \quad \overset{\text{KL divergence}}{\longleftrightarrow} \quad p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \tilde{\beta}_t \mathbf{I})$$

$$\Rightarrow \mathbb{E}_q[\|\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|^2]$$

Background: DDPM

$$\mathbb{E}_q[D_{KL}(q(\mathbf{x}_T | \mathbf{x}_0) \| p_\theta(\mathbf{x}_T)) + \sum_{t=2}^T D_{KL}(q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t)) - \log p_\theta(\mathbf{x}_0 | \mathbf{x}_1)]$$

what we're interested in

$$\Rightarrow q(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_{t-1}; \tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0), \tilde{\beta}_t \mathbf{I}) \quad \overset{\text{KL divergence}}{\longleftrightarrow} \quad p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \tilde{\beta}_t \mathbf{I})$$

$$\Rightarrow \mathbb{E}_q[\|\tilde{\mu}(\mathbf{x}_t, \mathbf{x}_0) - \mu_\theta(\mathbf{x}_t, t)\|^2]$$

Predicting denoised images is the same as predicting noise itself.

$$\Rightarrow \mathbb{E}_{\mathbf{x}_0, \epsilon}[\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2] \quad \mu_\theta(\mathbf{x}_t, t) = \frac{1}{\sqrt{\alpha_t}}(\mathbf{x}_t - \frac{\beta_t}{\sqrt{1 - \bar{\alpha}_t}} \epsilon_\theta(\mathbf{x}_t, t))$$

DDPM Inference

We train a noise predictor $\epsilon_\theta(\mathbf{x}_t, t)$, and the reverse pass is defined as:

$$p_\theta(\mathbf{x}_{t-1} | \mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \tilde{\beta}_t \mathbf{I}).$$

Sampling

```
 $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$   
for  $t = T, \dots, 1$  do  
   $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = 0$   
   $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \mu_\theta + \tilde{\beta}_t \mathbf{z}$   
end for  
return  $\mathbf{x}_0$ 
```

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We train a noise predictor $\epsilon_{\theta}(\mathbf{x}_t, t)$, and the reverse pass is defined as:

$$p_{\theta}(\mathbf{x}_{t-1} | \mathbf{x}_t) := \mathcal{N}(\mathbf{x}_{t-1}; \mu_{\theta}(\mathbf{x}_t, t), \tilde{\beta}_t \mathbf{I}).$$

Sampling

```
 $\mathbf{x}_T \sim \mathcal{N}(0, \mathbf{I})$   
for  $t = T, \dots, 1$  do too slow  
   $\mathbf{z} \sim \mathcal{N}(0, \mathbf{I})$  if  $t > 1$ , else  $\mathbf{z} = 0$   
   $\mathbf{x}_{t-1} = \frac{1}{\sqrt{\alpha_t}} \mu_{\theta} + \tilde{\beta}_t \mathbf{z}$   
end for  
return  $\mathbf{x}_0$ 
```


Reducing Inference Timesteps

Let's consider the forward process as defined not on all latents $\mathbf{x}_{1:T}$, but on a subset $\{\mathbf{x}_{\tau_1}, \dots, \mathbf{x}_{\tau_s}\}$.

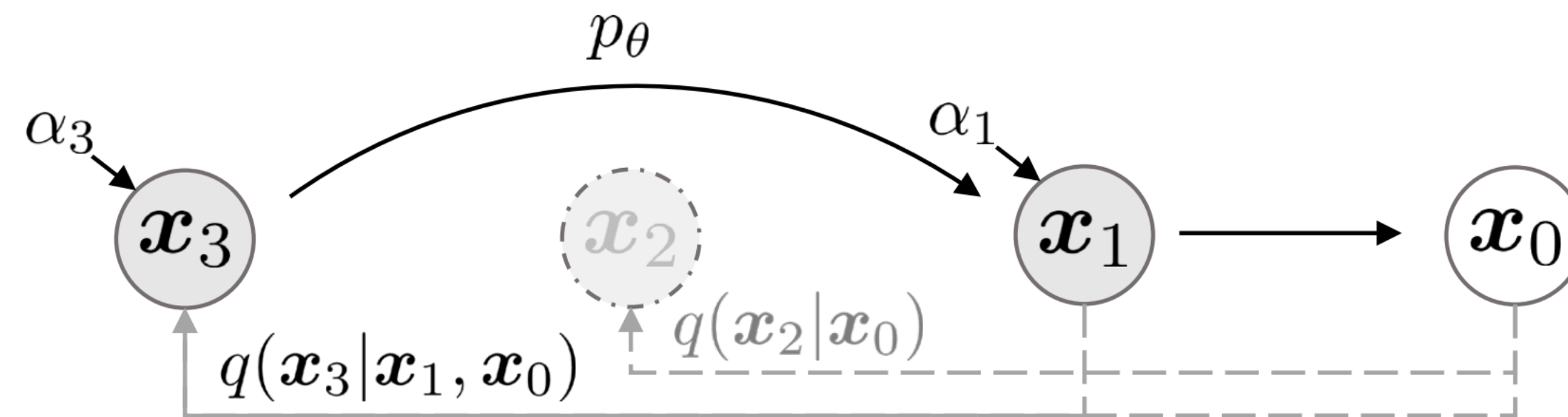


Figure 2: Graphical model for accelerated generation, where $\tau = [1, 3]$.

Reducing Inference Timesteps

Timesteps		CIFAR10 (32×32)					CelebA (64×64)				
	S	10	20	50	100	1000	10	20	50	100	1000
DDPM	0.0	13.36	6.84	4.67	4.16	4.04	17.33	13.73	9.17	6.53	3.51
	0.2	14.04	7.11	4.77	4.25	4.09	17.66	14.11	9.51	6.79	3.64
	0.5	16.66	8.35	5.25	4.46	4.29	19.86	16.06	11.01	8.09	4.28
	1.0	41.07	18.36	8.01	5.78	4.73	33.12	26.03	18.48	13.93	5.98
	$\hat{\sigma}$	367.43	133.37	32.72	9.99	3.17	299.71	183.83	71.71	45.20	3.26

The fewer timesteps, the less performance.
 $\Rightarrow$ What if we can control the stochastic?

Controlling Stochastic

We reformulate $q(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{x}_0)$ such that its stochastic is controllable by σ .

$$q(\mathbf{x}_t | \mathbf{x}_0) := \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0, (1 - \bar{\alpha}_t) \mathbf{I})$$

$$\implies \mathbf{x}_t = \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t} \epsilon_t$$

$$= \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t - \sigma_{t+1}^2} \epsilon_{t+1} + \sigma_{t+1} \epsilon$$

$$= \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t - \sigma_{t+1}^2} \frac{\mathbf{x}_{t+1} - \sqrt{\bar{\alpha}_{t+1}} \mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_{t+1}}} + \sigma_{t+1} \epsilon$$

$$q_\sigma(\mathbf{x}_t | \mathbf{x}_{t+1}, \mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t} \mathbf{x}_0 + \sqrt{1 - \bar{\alpha}_t - \sigma_{t+1}^2} \frac{\mathbf{x}_{t+1} - \sqrt{\bar{\alpha}_{t+1}} \mathbf{x}_0}{\sqrt{1 - \bar{\alpha}_{t+1}}}, \boxed{\sigma_{t+1}^2 \mathbf{I}})$$

DDIM Loss

$$\mathcal{J}_\sigma = \sum_t D_{KL}(q_\sigma(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_\theta^{(t)}(\mathbf{x}_{t-1} | \mathbf{x}_t))$$

where, $p_\theta^{(t)}(\mathbf{x}_{t-1} | \mathbf{x}_t) = q_\sigma(\mathbf{x}_{t-1} | \mathbf{x}_t, f_\theta(\mathbf{x}_t, t))$, $f_\theta(\mathbf{x}_t, t) := \frac{(\mathbf{x}_t - \sqrt{1 - \alpha_t} \cdot \epsilon_\theta(\mathbf{x}_t, t))}{\sqrt{\alpha_t}}$.

DDIM Loss

$$\mathcal{J}_\sigma = \sum_t D_{KL}(q_\sigma(\mathbf{x}_{t-1} | \mathbf{x}_t, \mathbf{x}_0) \| p_\theta^{(t)}(\mathbf{x}_{t-1} | \mathbf{x}_t))$$

$$\text{where, } p_\theta^{(t)}(\mathbf{x}_{t-1} | \mathbf{x}_t) = q_\sigma(\mathbf{x}_{t-1} | \mathbf{x}_t, f_\theta(\mathbf{x}_t, t)), \quad f_\theta(\mathbf{x}_t, t) := \frac{(\mathbf{x}_t - \sqrt{1 - \alpha_t} \cdot \epsilon_\theta(\mathbf{x}_t, t))}{\sqrt{\alpha_t}}.$$

Theorem 1. For all $\sigma > 0$, there exists $C \in \mathbb{R}$, such that $\mathcal{J}_\sigma = \mathcal{L} + C$.

$$\mathcal{L} = \sum_{t=1}^T \mathbf{E}_{\mathbf{x}_0, \epsilon} [\|\epsilon - \epsilon_\theta(\mathbf{x}_t, t)\|^2]$$

We can directly use a pre-trained DDPM as DDIM in inference time!

DDIM Inference

$$p_{\theta}^{(t)}(\mathbf{x}_{t-1} | \mathbf{x}_t) = q_{\sigma}(\mathbf{x}_{t-1} | \mathbf{x}_t, f_{\theta}(\mathbf{x}_t, t))$$

DDIM Inference

$$\begin{aligned} p_{\theta}^{(t)}(\mathbf{x}_{t-1} | \mathbf{x}_t) &= q_{\sigma}(\mathbf{x}_{t-1} | \mathbf{x}_t, f_{\theta}(\mathbf{x}_t, t)) \\ &= \mathcal{N}(\sqrt{\bar{\alpha}_T} f_{\theta}(\mathbf{x}_t, t) + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} f_{\theta}(\mathbf{x}_t, t)}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2 \mathbf{I}) \end{aligned}$$

DDIM Inference

$$\begin{aligned}p_{\theta}^{(t)}(\mathbf{x}_{t-1} | \mathbf{x}_t) &= q_{\sigma}(\mathbf{x}_{t-1} | \mathbf{x}_t, f_{\theta}(\mathbf{x}_t, t)) \\&= \mathcal{N}(\sqrt{\bar{\alpha}_T} f_{\theta}(\mathbf{x}_t, t) + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} f_{\theta}(\mathbf{x}_t, t)}{\sqrt{1 - \bar{\alpha}_t}}, \sigma_t^2 \mathbf{I}) \\ \mathbf{x}_{t-1} &= \sqrt{\bar{\alpha}_T} f_{\theta}(\mathbf{x}_t, t) + \sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2} \cdot \frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} f_{\theta}(\mathbf{x}_t, t)}{\sqrt{1 - \bar{\alpha}_t}} + \sigma_t \epsilon \\ \mathbf{x}_{t-1} &= \underbrace{\sqrt{\bar{\alpha}_T} f_{\theta}(\mathbf{x}_t, t)}_{\text{“prediction of } \mathbf{x}_0\text{”}} + \underbrace{\sqrt{1 - \bar{\alpha}_{t-1} - \sigma_t^2}}_{\text{“towards } \mathbf{x}_t\text{”}} \cdot \underbrace{\frac{\mathbf{x}_t - \sqrt{\bar{\alpha}_t} f_{\theta}(\mathbf{x}_t, t)}{\sqrt{1 - \bar{\alpha}_t}}}_{\text{“random noise”}} + \sigma_t \epsilon.\end{aligned}$$

Experiment Results

DDPM vs DDIM

With a small number of steps, DDIM outperforms DDPM by a significant margin.

Table 1: CIFAR10 and CelebA image generation measured in FID. $\eta = 1.0$ and $\hat{\sigma}$ are cases of **DDPM** (although [Ho et al. \(2020\)](#) only considered $T = 1000$ steps, and $S < T$ can be seen as simulating DDPMs trained with S steps), and $\eta = 0.0$ indicates **DDIM**.

	S	CIFAR10 (32×32)					CelebA (64×64)				
		10	20	50	100	1000	10	20	50	100	1000
DDIM	0.0	13.36	6.84	4.67	4.16	4.04	17.33	13.73	9.17	6.53	3.51
	0.2	14.04	7.11	4.77	4.25	4.09	17.66	14.11	9.51	6.79	3.64
	0.5	16.66	8.35	5.25	4.46	4.29	19.86	16.06	11.01	8.09	4.28
	1.0	41.07	18.36	8.01	5.78	4.73	33.12	26.03	18.48	13.93	5.98
DDPM	$\hat{\sigma}$	367.43	133.37	32.72	9.99	3.17	299.71	183.83	71.71	45.20	3.26

Sample Consistency in DDIM

From the same initial $\mathbf{x}_T$, different inference trajectories provide consistent results.

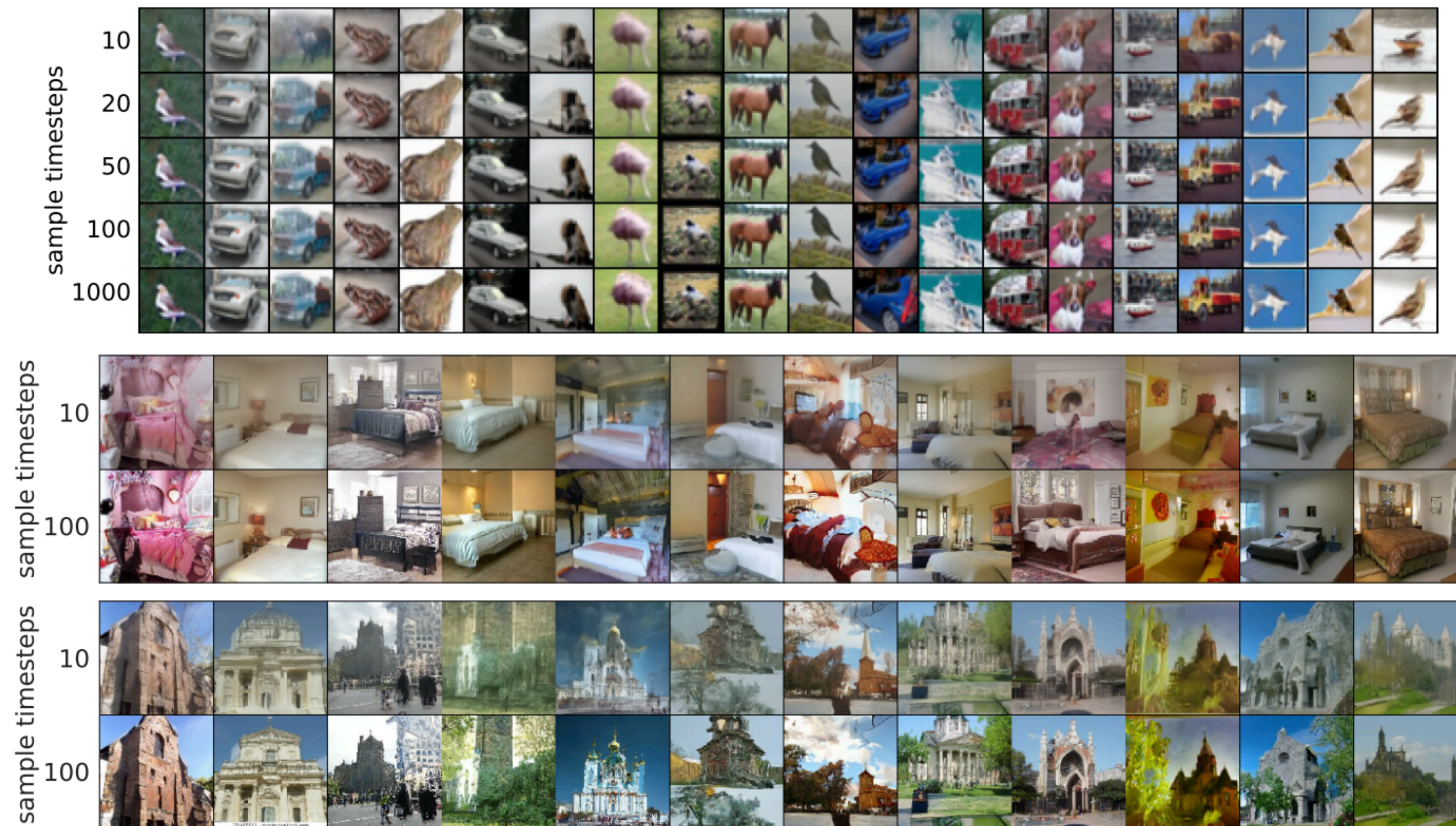


Figure 5: Samples from DDIM with the same random $\mathbf{x}_T$ and different number of steps.

Interpolation in DDIM

Deterministic process presents meaning interpolation in $\mathbf{x}_T$ space.

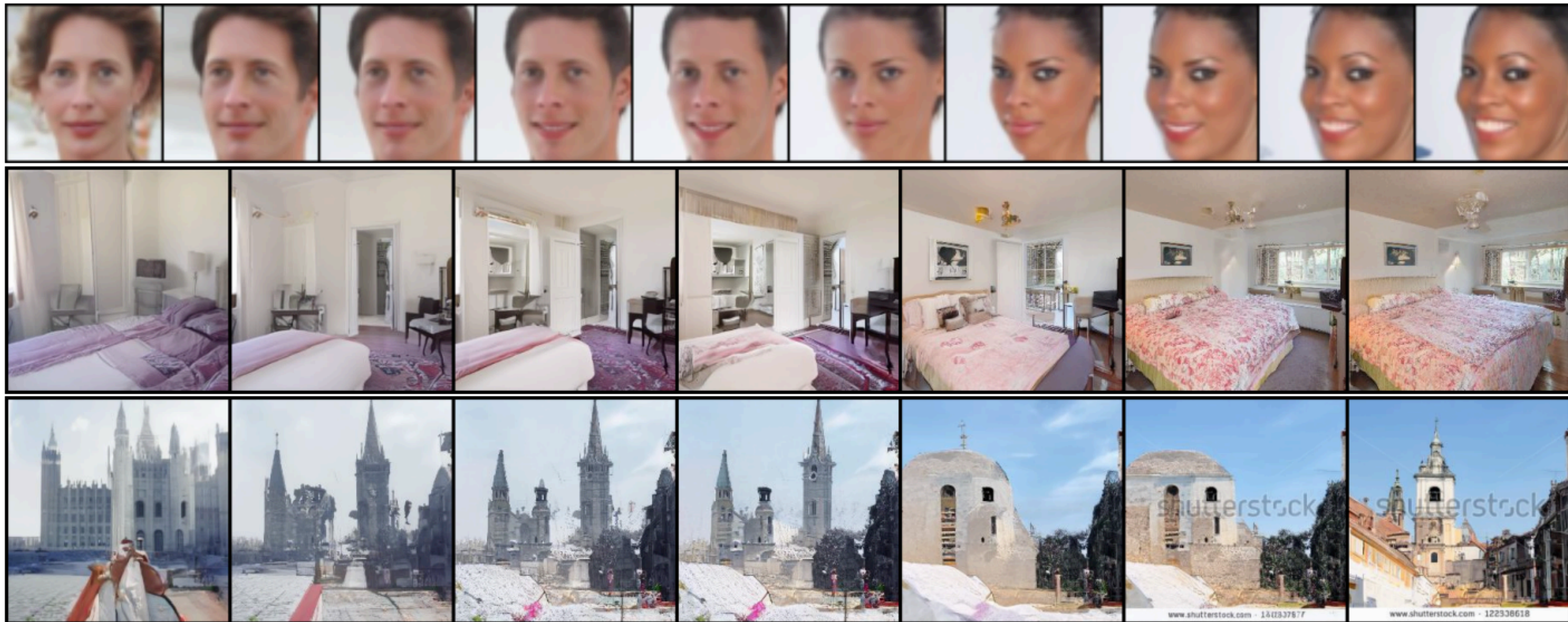


Figure 6: Interpolation of samples from DDIM with $\dim(\tau) = 50$.

**THANK YOU
FOR LISTENING**

The End.